

Unit 3.3 Trig derivatives

Note Title

2015/03/23

First two limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$s^2 + c^2 = 1$$

$$s^2 = 1 - c^2$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \times \frac{\cos x + 1}{\cos x + 1}$$

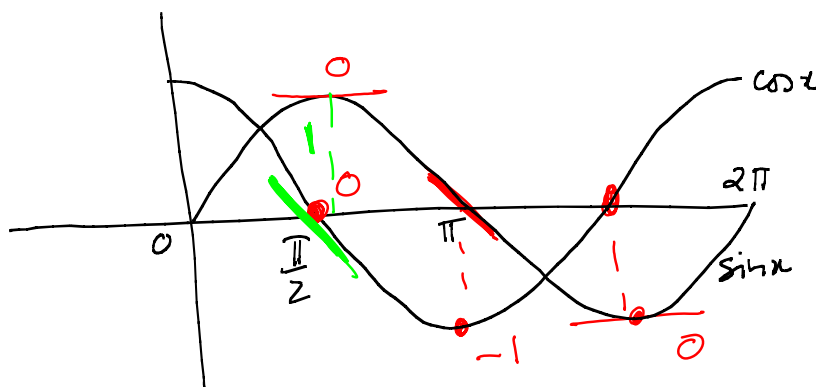
$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x(\cos x + 1)}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x(\cos x + 1)}$$

$$= \lim_{x \rightarrow 0} \underbrace{\frac{\sin x}{x}}_1 \cdot \underbrace{\sin x}_0 \cdot \underbrace{\frac{1}{\cos x + 1}}_{\frac{1}{2}}$$

$$= 0$$

Now: $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$



Slope of sin is the function value of cos.

NB:

We prove that $\frac{d}{dx} \cos x = -\sin x$

$$\frac{d}{dx} \cos x = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos x (\cos h - 1)}{h} - \lim_{h \rightarrow 0} \sin x \left(\frac{\sin h}{h} \right)$$

$$= \cos x \cdot 0 - \sin x \cdot 1$$

$$= -\sin x$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cdot \cot x$$

$$\frac{d}{dx} \sec x = \sec x \cdot \tan x$$

$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x.$$

Examples

$$1. \frac{d}{dt} (\sqrt{t} + \sin t) = \frac{1}{2} t^{-1/2} + \cos t$$

$$2. \frac{d}{dx} (x^3 \cdot \tan x) = 3x^2 \cdot \tan x + x^3 \cdot \sec^2 x$$

$$3. \frac{d}{dz} (2 \cos z \cdot \sec z) = -2 \sin z \cdot \sec z + 2 \cos z \cdot \sec z \cdot \tan z$$

$$4. \frac{d}{du} \left(\frac{e^u - 4 \cot u}{4 \operatorname{cosec} u} \right) +$$

$$= \frac{(e^u + 4 \operatorname{cosec}^2 u) 4 \operatorname{cosec} u + (e^u - 4 \cot u) 4 \operatorname{cosec} u \cdot \cot u}{(4 \operatorname{cosec} u)^2}$$

Why is $\frac{d}{dx} \tan x = \sec^2 x$?

$$\frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x.$$

Why is $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cdot \cot x$

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{\sin x} \right) &= \frac{-\cos x}{\sin^2 x} \\ &= -\frac{1}{\sin x} \times \frac{\cos x}{\sin x}. \end{aligned}$$

$$\left| \begin{aligned} \frac{d}{dx} \frac{1}{g(x)} \\ &= \frac{-g'(x)}{(g(x))^2} \end{aligned} \right|$$